LESSON PLAN

GAS TURBINE ENGINE (16)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Eddy current inspection, principle of operation, function of basic system, coil impedance, electrical conductivity mag permeability, types of inspection coils, application of various types of parts
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Eddy current inspection, principle of operation, function of basic system, coil impedance, electrical conductivity mag permeability, types of inspection coils, application of various types of parts
5. Specific Objectives	16.1.1.1 Trainee will be able to describe types of NDT and perform liquid penetrant, magnetic particles and eddy current inspections

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Eddy Current testing on test blocks using various probe types; Safety precautions regarding high voltage and equipment handling; Instructor supervision during probe coupling and signal interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three main types of NDT methods?
(b) Which NDT method is used for surface crack detection in non-ferrous metals?
(c) What is the primary difference between magnetic particle and eddy current inspection?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of operation of eddy current inspection • Function of basic eddy current system components • Concept of coil impedance and its variation • Role of electrical conductivity and magnetic permeability • Types of inspection coils and their configurations	L & Ds	PPT, IB	15 min
(b)	• Application of various coil types to specific parts • Interpretation of impedance plane diagrams • Factors affecting test sensitivity and resolution • Calibration procedures for eddy current instruments • Standard test blocks and defect simulation techniques	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting common eddy current test failures • Safety protocols for high frequency electrical systems • Maintenance of eddy current probes and cables • Documentation of inspection results and reports • Comparison of eddy current with other NDT methods	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how magnetic permeability affects the depth of penetration in eddy current testing.
(b) Describe the function of the reference coil in an eddy current instrument.
(c) Why is electrical conductivity a critical factor in selecting test parameters?

12. Home Assignment and Follow Up	Prepare a brief summary comparing the applicability of eddy current inspection versus radiographic inspection for detecting subsurface defects in aluminum alloy engine components, referencing the next lesson plan on Radiographic inspection.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Radiographic inspection, use, applicability inspection principle of radiography, X-ray gamma ray, shadow for motion, types of recording, types of test specimen safety precaution while using x-ray	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min